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SUMMARY

Senior Data Engineer with 20 years of experience, including 4 years at Amazon (Alexa Shopping, JWO) architecting and owning petabyte-scale data platforms. Deep specialization in large-scale Apache Spark, delta lake architectures (Iceberg, Hudi, Delta Lake), ML/LLM data infrastructure on AWS, and privacy-compliant data systems. Track record of end-to-end ownership across streaming and batch systems, with cross-team technical influence and a prior people-leadership engagement at Capital One and cognizant.

SKILLS

- **Languages & Frameworks:** Python, Scala, Java, SQL, TypeScript, Shell
- **Big Data & Processing:** Apache Spark (PySpark/Scala/Java), Databricks, Kafka, Kinesis, Apache Iceberg, Apache Hudi, Delta Lake
- **AWS Data Stack:** S3, EMR, Glue, Redshift, Athena, Kinesis (Streams + Firehose), Lambda, Step Functions, SageMaker, Bedrock, DMS, AWS CDK
- **Databases & Warehouses:** Redshift, Snowflake, DynamoDB, DocumentDB, RDS
- **ML / AI Infrastructure:** Bedrock (Claude Sonnet 4.5), SageMaker, spaCy, scikit-learn, model serving and evaluation pipelines
- **Workflow & Orchestration:** AWS Step Functions, Apache Airflow
- **Architecture & Practices:** Delta lake architectures, large-scale schema design, ETL/ELT, real-time + batch hybrid systems, data privacy & governance (GDPR), feature stores

EXPERIENCE

DATA ENGINEER

Amazon, May 2022– Present

- **Customer Data Deletion Platform (Alexa Shopping):** Architected and owned hybrid streaming + batch GDPR-compliant deletion system on Amazon's centralized data lake. Kinesis streaming layer handles 4k TPS for request ingestion and acknowledgement; Spark batch layer processes 4 PB to mark records for deletion by updating retention metadata (DMA tagging framework) on raw S3 partitions, integrating with Amazon's centralized retention-driven purge pipeline under privacy team governance.
- **PII Detection System (Alexa Shopping):** Architected and own ML-based PII detection system scanning 25 TB across the data lake, using Microsoft Presidio's Spacy NLP engines to identify sensitive customer data.
- **ML/LLM Data and Annotation Platform (Alexa Shopping):** Architected and own end-to-end data and inference platform for customer or shopping intent and mission understanding models for annotation platform. Manage data

pipelines (training data, features, downstream output consumption) and production Bedrock invocation infrastructure for classification models and Claude Sonnet 4.5, serving tens of thousands of daily invocations.

- **Org-Level Feature Store POC (Alexa Shopping):** Led organization-wide POC evaluating Apache Iceberg, Delta Lake, and Apache Hudi for building a feature store; recommendation of Iceberg was adopted as the org standard. Modified an internal Amazon data platform to integrate with Iceberg security model; modified code reused by multiple downstream teams.
- **Near Real-Time Analytics Delivery Platform (Just walk out store):** Architected and own end-to-end analytics platform extracting data from Hudi-based delta lake and Redshift, delivering refreshed insights into 200+ customer-specific QuickSight dashboards.
- **Config driven ETL framework:** Architected and own the ETL framework which can source the data from multiple sources (eg: S3, salesforce, ServiceNow, smart sheets) and ingest it into AWS JWO's data warehouse. The config driven ETL framework, reduced the development and operational effort from 40 hours to 1 hour.
- **Cross-team Spark Consulting (Alexa Shopping):** Recognized go-to engineer for Apache Spark scaling problems across 5+ teams in Alexa Shopping; helped teams diagnose and fix performance, cost, and reliability issues on data pipelines running at production scale.

CONSULTANT – LEAD DATA ENGINEER

Capital One Financial Services, April 2021–May 2022

- Led 10-engineer team building end-to-end Spark-based ML feature engineering pipeline for customer address matching across millions of records; pipeline drove customer record deduplication and customer splitting (single customer view) for downstream banking applications. Sourced data from Cassandra, processed features in Spark (Java), and delivered analytics, reports, and alarms via Databricks.
- Owned team leadership including interviewing, hiring, and onboarding of consultant engineers; managed customer stakeholder collaboration across the engagement.

ARCHITECT

Cognizant Technology Solutions, August 2012–April 2021

- Lead Spark/Hadoop engineer on mainframe modernization program for a US electric utility client. Built reusable, config-driven ETL framework migrating data from mainframe DB2 and ADABAS via Sqoop into Parquet on HDFS, with Spark-based transformations landing into Hive and Impala for analytics. Framework handled ingestion across many source tables, replacing legacy mainframe ETL.
- Earlier in tenure, worked across database engineering and big data roles spanning DB2 z/OS administration, SQL performance tuning, and progressive transition into Hadoop, Spark, AWS (S3, EMR, Redshift, Glue), and streaming (Kafka, Kinesis).
- Owned team leadership including interviewing, hiring, and onboarding of consultant engineers; managed customer stakeholder collaboration across the engagement.

Earlier Experience (2006–2012) — DB2 z/OS database administration, SQL performance tuning, and database automation roles at HCL Technologies and Fidelity National Information Services

EDUCATION

Post Graduate Program in Big Data Engineering — BITS Pilani, 2018

Master of Computer Applications (MCA) — Gandhigram Rural Institute, 2006

B.Sc. Computer Science — Kannur University, 2002

CERTIFICATIONS

- AWS certified BIG Data - Specialty Certification
- AWS Certified Solution Architect